

Lecture: 6

Percentage



Percentage

$$\% \Rightarrow 6\% = \frac{6}{100}$$

Out of 100 $\rightarrow$ %

mm $\rightarrow$ 100, 60m

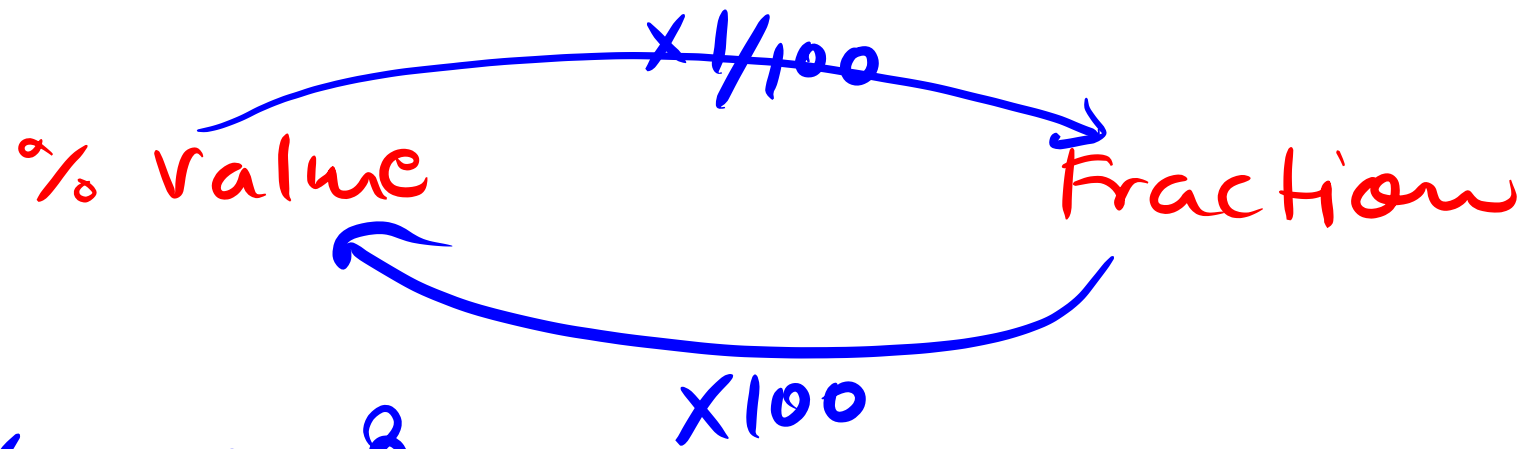
$$\begin{array}{r|l} \text{mm} \rightarrow & \boxed{200, 80} \\ & \underline{\quad} \quad \underline{\quad} \\ & 2 \quad \quad 2 \end{array}$$

$$\begin{array}{r} 100 \quad 40 \\ \hline 40\% \end{array}$$

$$\begin{array}{r} 60\% \\ \hline 80^2 \\ \hline 200 \\ 5 \end{array}$$

$$\begin{array}{r} 215 \times 100 \\ \hline = 40\% \end{array}$$





$$6\% = \frac{6}{100}$$

$$\frac{2}{5} = \frac{2}{5} \times 100 = \underline{\underline{40\%}}$$



$$1 = 100\%$$

$$\frac{1}{2} = 50\%$$

$$\frac{1}{3} = 33.33\%$$

$$\frac{1}{4} = 25\%$$

$$\frac{1}{5} = 20\%$$

$$\frac{1}{6} = 16.66\%$$

$$\frac{1}{7} = 14.28\%$$

$$\frac{1}{8} = 12.5\%$$

$$\frac{1}{9} = 11.11\%$$

$$\frac{1}{10} = 10\%$$

IMP Fraction

$$\frac{1}{11} = 9.09\%$$

$$\frac{1}{12} = 8.33\%$$

$$\frac{1}{13} = 7.69\%$$

$$\frac{1}{14} = 7.14\%$$

$$\frac{1}{15} = 6.66\%$$

$$\frac{1}{16} = 6.25\%$$

$$\frac{1}{17} = 5.88\%$$

$$\frac{1}{18} = 5.55\%$$

$$\frac{1}{19} = 5.26\%$$

$$\frac{1}{20} = 5\%$$



$$40\% \rightarrow 2 \times 20\%$$

$$2 \times 1/5 = \textcircled{2/5}$$

$$133.33\% \Rightarrow 100\% + 33.33\%$$

$$\rightarrow 4 \times 33.33\% \quad 1 + 1/3 = \textcircled{4/3}$$

$$4 \times 1/3 = \textcircled{4/3}$$

$$\underline{62.5\%}$$

$$50\% + 12.5\%$$

$$1/2 + 1/8 = \textcircled{5/8}$$



266.67%

160%

$$2 + \frac{2}{3} = \underline{\underline{\left(\frac{8}{3}\right)}}$$

450%

$$4 + \frac{1}{2} = \underline{\underline{\left(\frac{9}{2}\right)}}$$

87.5%

$$75\% + 12.5\%$$

$$\frac{3}{4} + \frac{1}{8}$$

$$= \underline{\underline{\left(\frac{7}{8}\right)}}$$

$$100\% - 12.5\% = 1 - \frac{1}{8}$$

$$= \underline{\underline{\left(\frac{7}{8}\right)}}$$

$$7 \times 12.5\%$$

$$= \underline{\underline{\left(\frac{7}{8}\right)}}$$





Contains:

- Calculation of percentage
- Percentage Comparison
- Percentage to Fraction conversion
- Problem based on marks





Percentage: Per means divided and cent means 100.

To convert any value on the base of hundred.

Conversion of percentage to fraction: To convert percentage to fraction, divide that percent value by 100.

E.G. $50\% = 50/100 = \frac{1}{2}$

$$25\% = 25/100 = \frac{1}{4}$$

Conversion of fraction to percentage: To convert fraction to percentage multiply the fraction by 100.

For e.g. $\frac{3}{8} = (\frac{3}{8}) * 100 = 37.5\%$



$\frac{1}{2} = 50\%$			
$\frac{1}{3} = 33\frac{1}{3}\%$	$\frac{2}{3} = 66\frac{2}{3}\%$		
$\frac{1}{4} = 25\%$	$\frac{3}{4} = 75\%$		
$\frac{1}{5} = 20\%$	$\frac{2}{5} = 40\%$	$\frac{3}{5} = 60\%$	$\frac{4}{5} = 80\%$
$\frac{1}{6} = 16\frac{2}{3}\%$	$\frac{5}{6} = 83\frac{1}{3}\%$		
$\frac{1}{7} = 14\frac{2}{7}\%$	$\frac{2}{7} = 28\frac{4}{7}\%$	$\frac{3}{7} = 42\frac{6}{7}\%$	$\frac{4}{7} = 57\frac{1}{7}\%$
		$\frac{5}{7} = 71\frac{3}{7}\%$	$\frac{6}{7} = 85\frac{5}{7}\%$
$\frac{1}{8} = 12\frac{1}{2}\%$	$\frac{3}{8} = 37\frac{1}{2}\%$	$\frac{5}{8} = 62\frac{1}{2}\%$	$\frac{7}{8} = 87\frac{1}{2}\%$
$\frac{1}{9} = 11\frac{1}{9}\%$	$\frac{2}{9} = 22\frac{2}{9}\%$	$\frac{4}{9} = 44\frac{4}{9}\%$	$\frac{5}{9} = 55\frac{5}{9}\%$
		$\frac{7}{9} = 77\frac{7}{9}\%$	$\frac{8}{9} = 88\frac{8}{9}\%$
$\frac{1}{10} = 10\%$	$\frac{3}{10} = 30\%$	$\frac{7}{10} = 70\%$	$\frac{9}{10} = 90\%$
$\frac{1}{11} = 9\frac{1}{11}\%$	$\frac{2}{11} = 18\frac{2}{11}\%$	$\frac{3}{11} = 27\frac{3}{11}\%$	$\frac{4}{11} = 36\frac{4}{11}\%$
$\frac{1}{12} = 8\frac{1}{3}\%$			





Let's Practice.

$$5 \times 11.11\% = \frac{5}{9}$$

Question : Find 55.55 % of 45.

$$\frac{5}{9} \times 45 = 25$$

$$\frac{5}{11} \times 44 = 20$$

Question : Find 45.45 % of 44

$$5 \times 9.09\%$$



Conversion Percentage to Fraction

1. Convert 62.5% to a fraction.

$$5/8$$

2. Convert 28.56 % to a fraction.

$$2 \times 14 = 28\% \\ 2 \times 1/7 = \frac{2}{7}$$





3. Convert 130% to a fraction.

$$\frac{13}{10}$$

4. Convert 460% to a fraction.

$$\frac{460}{100} = \frac{23}{5}$$

5. Convert 133.33% to a fraction.

$$\frac{4}{3}$$

① $a\% \text{ of } b = \frac{a \times b}{100}$

② $a \text{ is what \% of } b$
 $= \frac{a}{b} \times 100$

③ $a\% \text{ of } b = b\% \text{ of } a$





Q. If 16% of 40% of a number is 8, then find the number.

A. 120

B. 125

C. 130

D. 135

$$\frac{16}{100} \times \frac{40}{100} \times N = 8$$
$$N = 125$$



Q. What percentage of $\frac{2}{7}$ is $\frac{1}{35}$?

- A. 10%
B. 20%
C. 25%
D. none

$$\frac{\frac{1}{35} \times 100}{\frac{2}{7}}$$

$$\frac{1}{35} \times \frac{7}{2} \times 100$$

$$= 10\%$$



Q. 16.66% of 300.

A. 50

B. 75

C. 30

D. 18.88

$$\rightarrow \frac{1}{6} \times 300 = 50$$

45% of 133.33 =

133.33% of 45

$$\frac{4}{3} \times 45 = \underline{60}$$

$$10\% = \frac{1}{10}$$

$$10 = \frac{1}{10}$$





$X \% \text{ of } Y = Y\% \text{ of } X$

Question : Find 45% of 133.33

Answer : 45% of 133.33 = 133.33 % of 45

$$= (100+33.33)\% \text{ of } 45$$

$$= (1+1/3) * 45$$

$$= 4/3 * 45$$

$$= 60$$





Percentage Comparison

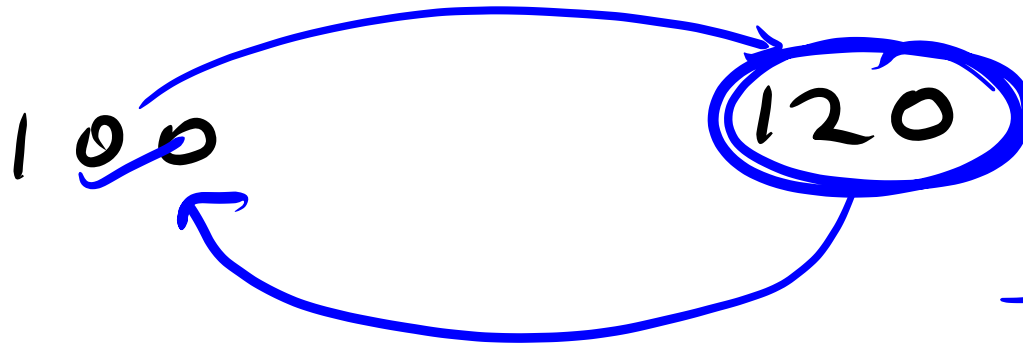
- 40 is what percentage of 60?
- ~~60~~ is how much percentage greater than 40?
- ~~40~~ is how much percentage lesser than 60?

$$\frac{40}{60} \times 100 = \underline{\underline{66.66\%}}$$

$$\frac{20}{40} \times 100 = 50\%$$

$$\frac{20}{60} \times 100 = \underline{\underline{33.33\%}}$$

20% ↑, 0% = ?



$$\frac{20}{120} \times 100 = \underline{16.6\bar{6}\%}$$





Less

Question : If A's salary is 20 % ~~more~~ than B, then by how much Percentage B's salary is ~~less~~ than A.

More

(a) 20%

(b) 25%

(c) 16.66%

(d) 30%

$\frac{1}{5} \rightarrow \frac{1}{4}$

25%



Question : If A's marks in an exam is 40 % less than B then by how much % B's marks are more than A.

(a) 40 %

(a) 60 %

(a) 66.66 %

(a) 20 %

$$\begin{array}{l} B \rightarrow 100 \\ A \rightarrow 60 \end{array} \quad \leftarrow 40$$

$$\frac{40}{60} \times 100 = \underline{\underline{66.67\%}}$$

$$\frac{2}{5} \downarrow$$

$$\frac{2}{5-2} = \frac{2}{3}$$

$$\downarrow$$
$$\underline{\underline{66.66\%}}$$



Question : If $16\frac{2}{3}\%$ of a number is added to itself the number becomes 700. Find original number.

$$x + \frac{1}{6}xx = 700$$

$$\frac{7x}{6} = 700$$

$x = 600$ Ans

(a) 400

(a) 600

(a) 800

(a) 700



Question: A student multiplied a number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the percentage error in the calculation?

(a) 44%

(b) 64%

(c) 40%

(d) 60%

$$15 \times \frac{3}{5} = 9 \quad \text{false error}$$

$$15 \times \frac{5}{3} = 25 \quad \text{True}$$

$$\frac{16}{25} \times 100 = 64\%$$



Question: The number was being multiplied by $\frac{5}{6}$. By mistake it is ~~divided by~~ $\frac{5}{6}$. Find percentage error in result.

(a) 44%

(b) 64%

(c) 40%

(d) 60%

$$\begin{array}{l} 30 \times \frac{5}{6} = \boxed{25} \quad \text{True} \\ 30 \times \frac{6}{5} = \boxed{36} \quad \text{False} \end{array}$$

$$\frac{11}{25} \times 100 = \underline{\underline{44\%}} \quad \text{Ans}$$



Problem based on Marks





Question : A student scored 140 Marks and still failed by 35 marks. If the passing criteria of that exam is 35%. Then find the maximum marks of that exam.

$$140 + 35 = 35\%$$

(a) 500

$$175 = 35\%$$

(a) 600

$$1\% = 5$$

(a) 1000

$$100\% = 5 \times 100 \\ = \underline{\underline{500}}$$

(a) 700



Question : A student scored 25 % in an examination and still failed by 30 marks while another candidate scored 50% marks and get 20 marks more than the passing marks. Then find the passing percentage. , mm , Pass marks

$$25\% + 30 = 50\% - 20$$

$$25\% = 50$$

$$1\% \rightarrow 2$$

$$25\% + 15\% = 40\%$$

$$50 + 30 = 80$$

$$100\% = 200$$

(a) 30 %

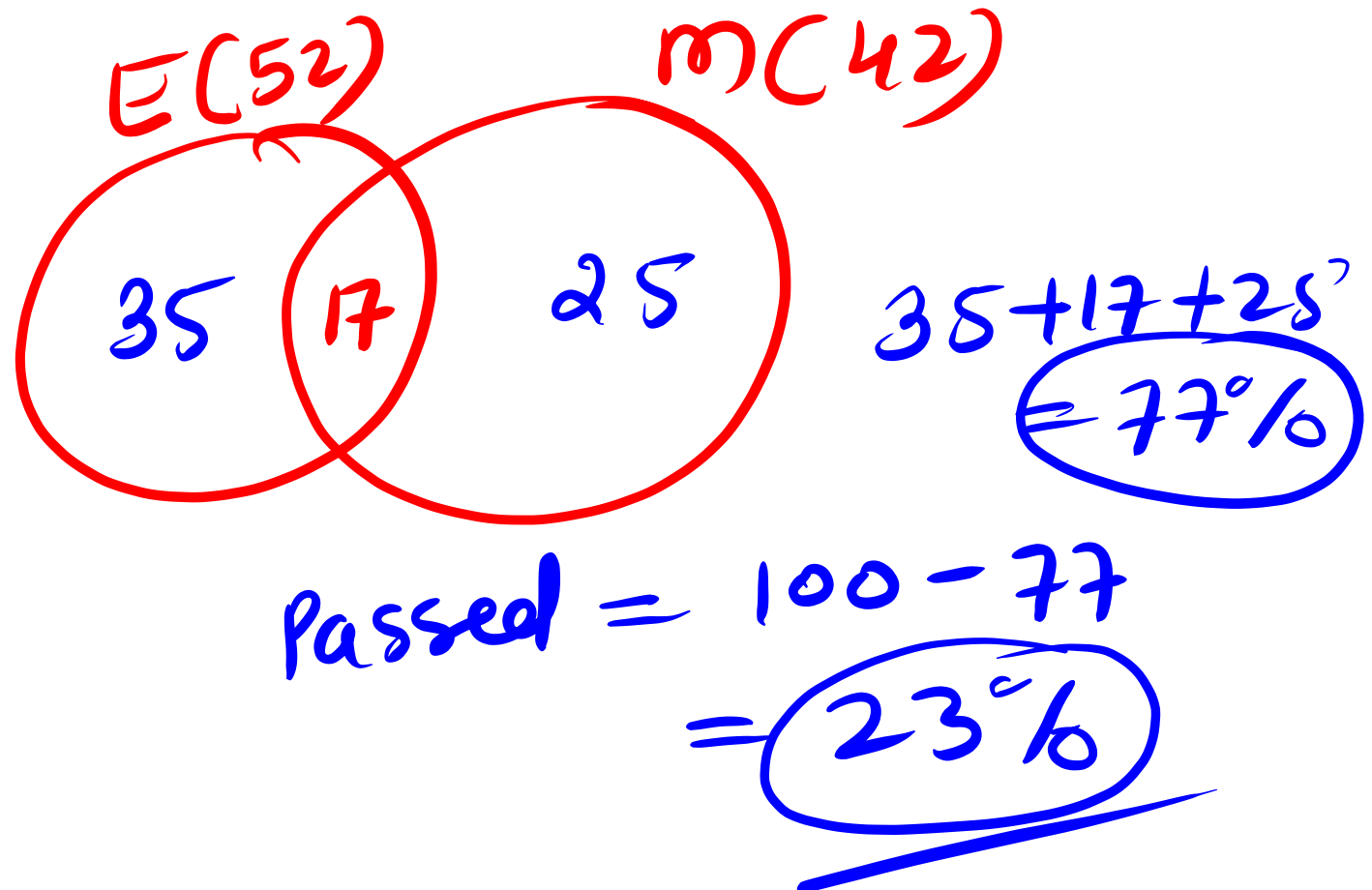
(a) 40 %

(a) 45 %

(a) 50 %



Question: In an Exam, 52% candidates failed in English, 42% in mathematics and 17% in both. What was the number of percentage of passed students in both subjects?

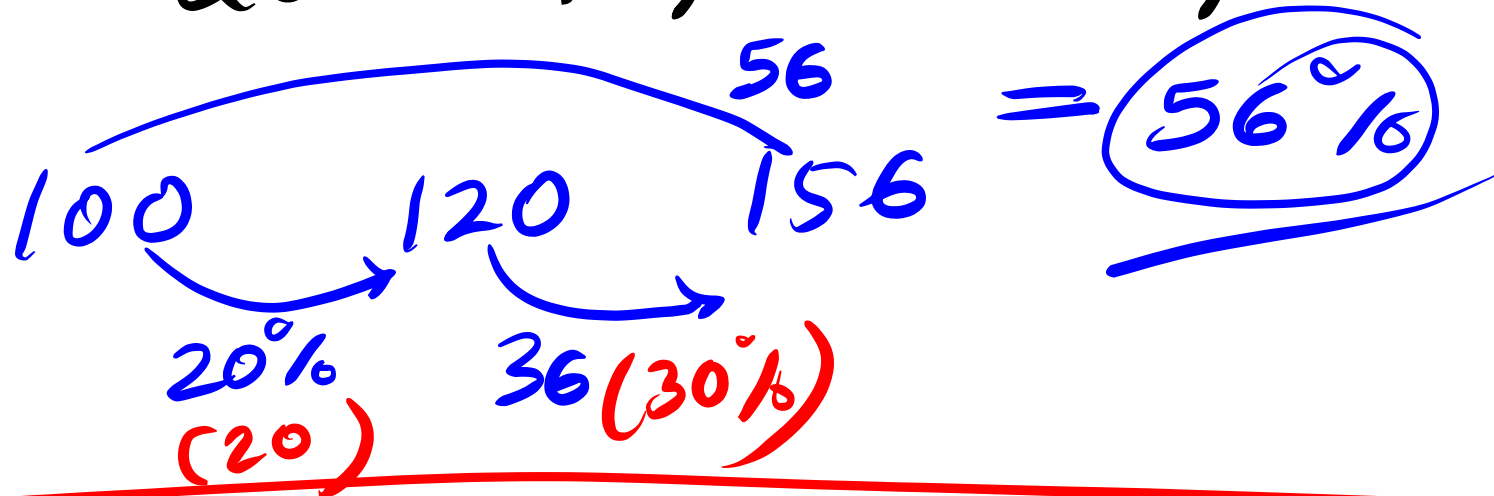


- (a) 23
- (b) 77
- (c) 6
- (d) 94



Successive %

Q 20% ↑, 30% ↑, Overall Int%?



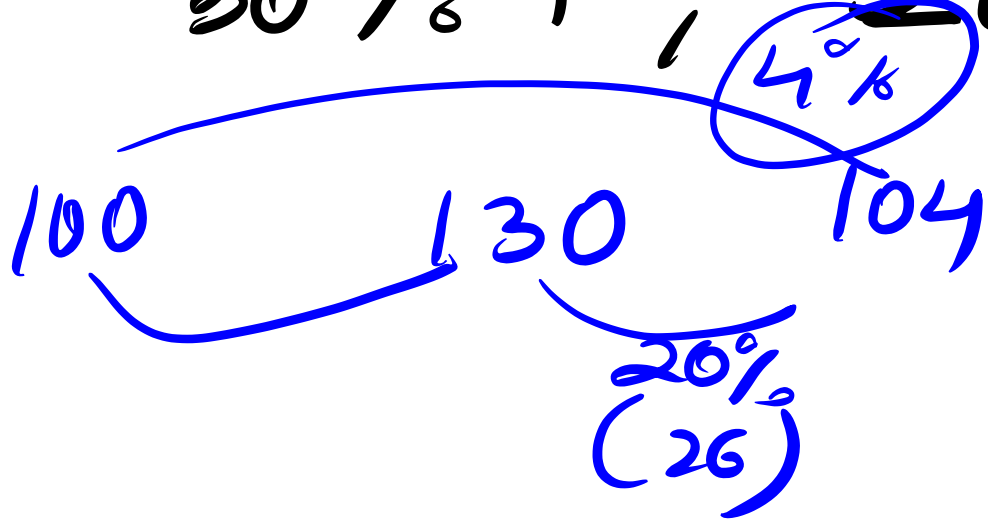
$$a + b + \frac{ab}{100}$$

$$20 + 30 + \frac{20 \times 30}{100}$$

Ans 56

(↑) → (+) Positive
(↓) → (-) Negative

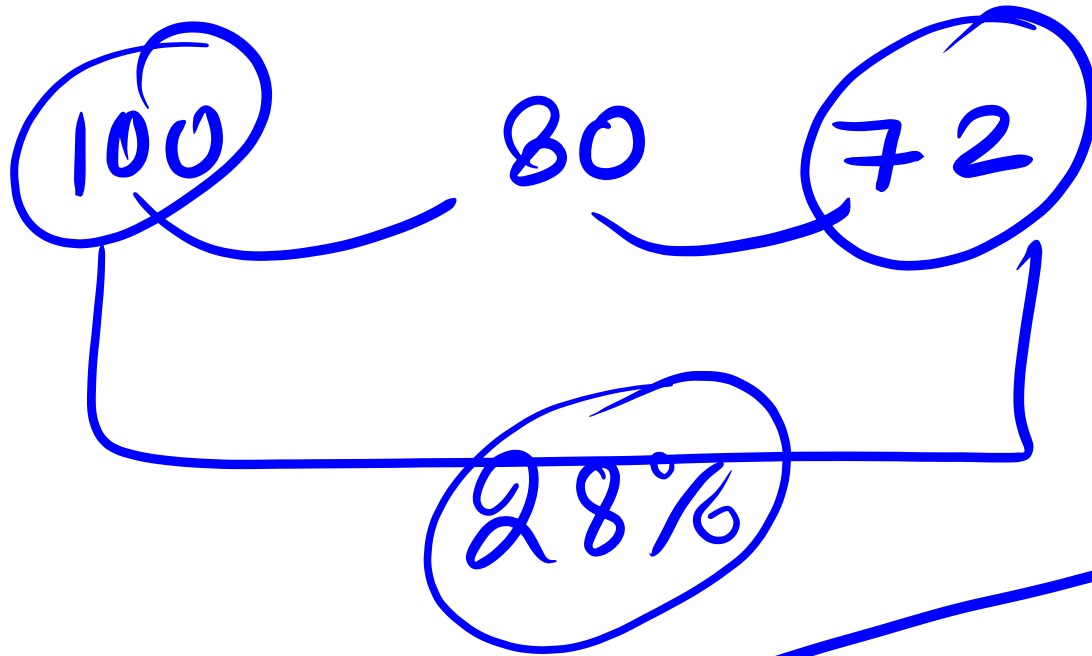
30% ↑, ~~30%~~ ↓, I/O %?



$$30 + (-20) + \frac{30 \times -20}{100}$$
$$10 - 6 = \underline{\underline{4\%}} =$$



20% ↓ , 10% ↓ , overall D%?



$$\begin{aligned} & \checkmark \quad \checkmark \\ & (-20) + (-10) \\ & + \frac{-20 \times -10}{100} \end{aligned}$$

$$-30 + 2$$

$$= \boxed{-28}$$

Decrease



Lecture: 7

Percentage





Concept to discuss

1. Successive Percentage Change.
2. Problem based on Elections.
3. Population change & percentage error.





Successive Increase and Decrease

$$\text{Net change} = a + b + \frac{ab}{100}$$

For Increase = Take positive sign

For decrease = Take negative sign





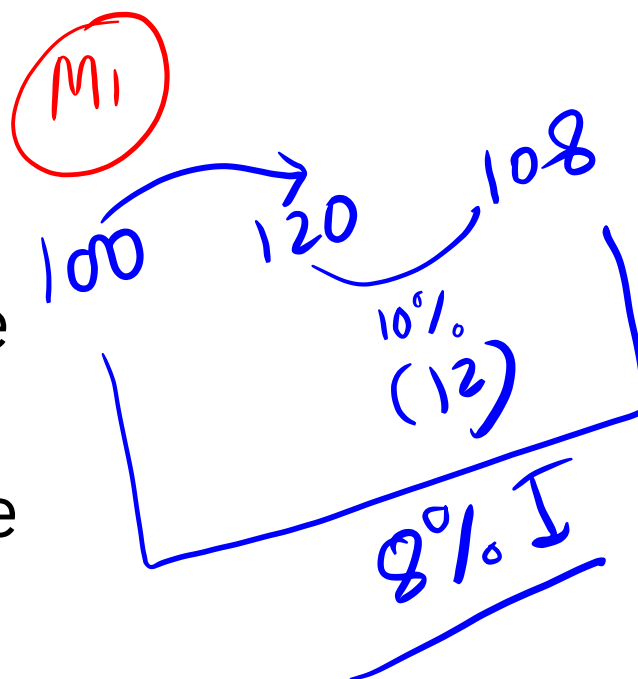
Question : Price of petrol first increased by 20% and then it is decreased by 10% . Find the net change in the price.

(a) ~~8 % increase~~

(a) 2 % decrease

(a) 10 % increase

(a) 8 % decrease



M_2

$$a + b + \frac{ab}{100}$$
$$20 + (-10) + \frac{20 \times -10}{100}$$
$$10 - 2 = 8\%$$



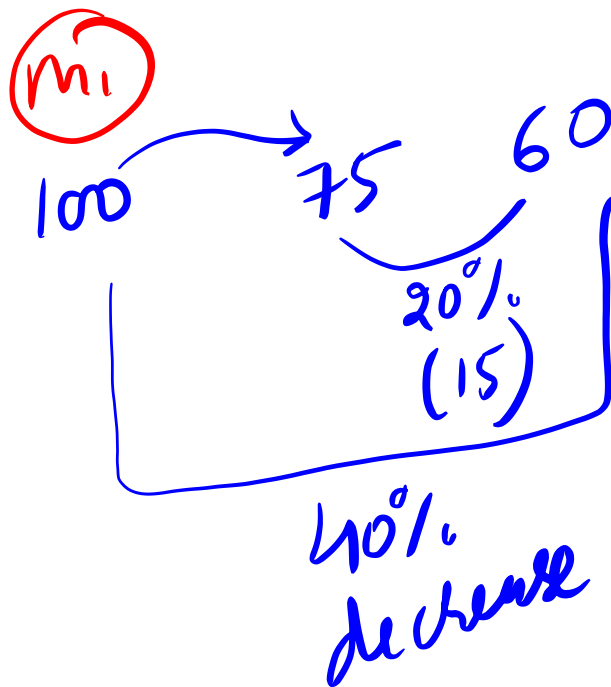
Question : Demand of a car went down by 25 % in 2016 and 20 % in 2017. What is net % decrease in demand?

(a) 45 %

(a) 40 %

(a) 50 %

(a) 60 %



M_2

$$(-25) + (-20) + \frac{-25 \times -20}{100}$$
$$= -45 + 5$$
$$= -40$$

40% decrease



Question : If the length of a rectangle is increased by 20 % and width is decreased by 30% then find the resultant change in area.

$$A = L \times B$$

~~(a) 16 %~~

(a) 20 %

(a) 24 %

(a) 28 %

$L = 10$
 $B = 10$
 $A = 100$

$\xrightarrow{20\%}$ $L(\text{New}) = 12$
 $\xrightarrow{30\%}$ $B(\text{New}) = 7$
 $A(\text{New}) = 84$

$\rightarrow 16\%$ decrease

(m2)

$$20 + (-30) + \frac{20 \times -30}{100}$$

$$-10 - 6$$

$$= -16$$

16% decrease



Question : A number is first increased by 15% and then decreased by 20%. The number so obtained is 64 less than the original number. What is the original number?_____

- (a) 600
- (a) 750
- (a) 800
- (a) 860

$$\begin{array}{ccc} 100 & 115 & 92 \\ & \underbrace{\quad\quad\quad} & \\ & 20\% (23) & \\ & \underbrace{\quad\quad\quad} & \\ & 8 & \end{array}$$

$8\text{ unit} \rightarrow 64$
 $1\text{ unit} = 8$
 $100\text{ unit} = 800$



Question : If the price of petrol is raised by 20% then the percentage by which a car owner must reduce his consumption so that there is no change in expenditure.

↳ In this scene, Price Inversely Proportional to Quantity / Consumption

(a) 16.66 %

(a) 18 %

(a) 15 %

(a) 25 %

Price	100	120
Quant	120	100

20

$$\frac{20}{120} \times 100 = 16.66\%$$



We know , **Expenditure = Price × Consumption**

Convert given % into fraction (a/n)

• If Increase $\frac{a}{n}$ $\longleftrightarrow$ Decrease $\frac{a}{n+a}$

For this case if $1/5$ increase ----- $1/6$ decrease

$$1/6 = 100/6 \% = 16.66 \%$$

• If Decrease $\frac{a}{n}$ $\longleftrightarrow$ Increase $\frac{a}{n-a}$

For this case if $2/5$ decrease ----- $2/3$ Increase





Question :If the price of sugar is decreased by 12.5% then the percentage by which one household must increase his consumption so that there is no change in expenditure.

- (a) 10 %
- (a) 8 %
- (a) 14.28 %
- (a) 12.5 %
- Price 100 87.5
Quant 87.5 100
12.5
12.5 x 100
87.5
= 14.28%



Question: If the price of sugar is increased by 25% then by how much percentage consumption should be reduced so that the expenditure will increase by only 5%

$$\text{Exp} = P \times Q \Rightarrow Q = \frac{\text{Exp}}{P}$$

$$P(\text{New}) = 125$$

$$\text{Exp} = 10500$$

$$Q(\text{New}) = \frac{10500}{125} = 84$$

16% reduction

let

$$P = 100$$

$$Q = 100$$

$$\text{Exp} = 100 \times 100 = 10000$$

(a) 25%

(b) 15%

~~(c) 16%~~

(d) 20%



Question : If the price of sugar is reduced by 20% due to which a person can buy 2kg more sugar for Rs. 200. Find the original price of sugar per kg.

Exp is same here

So, $P \propto \frac{1}{Q}$, Consumption = $P \times Q$

(a) ~~Rs. 25 per Kg~~

(a) Rs. 20 per Kg

(a) Rs. 22 per Kg

(a) Rs. 16 per Kg

Price	100	80
Qun	80	100
	4	5

↓
8kg
1 unit = 2kg → 10kg

original price = $\frac{\text{Exp}}{\text{original quantity}}$

⇒ $\frac{200}{8} = \boxed{25}$ Ans





Problem based on Elections





Question: In an Election between two candidates one got 65% of the votes and won by 300 votes. Find total no. of votes.

(a) 1200

(b) 1500

(c) 1800

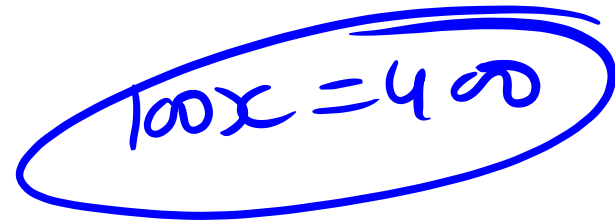
~~(d) 1000~~

$$\begin{array}{c} 100x \\ \swarrow \quad \searrow \\ 65x \quad 35x \\ \underbrace{\hspace{10em}} \\ 30x = 300 \\ \textcircled{x = 10} \end{array}$$

$$100x = 1000$$



(a) 600





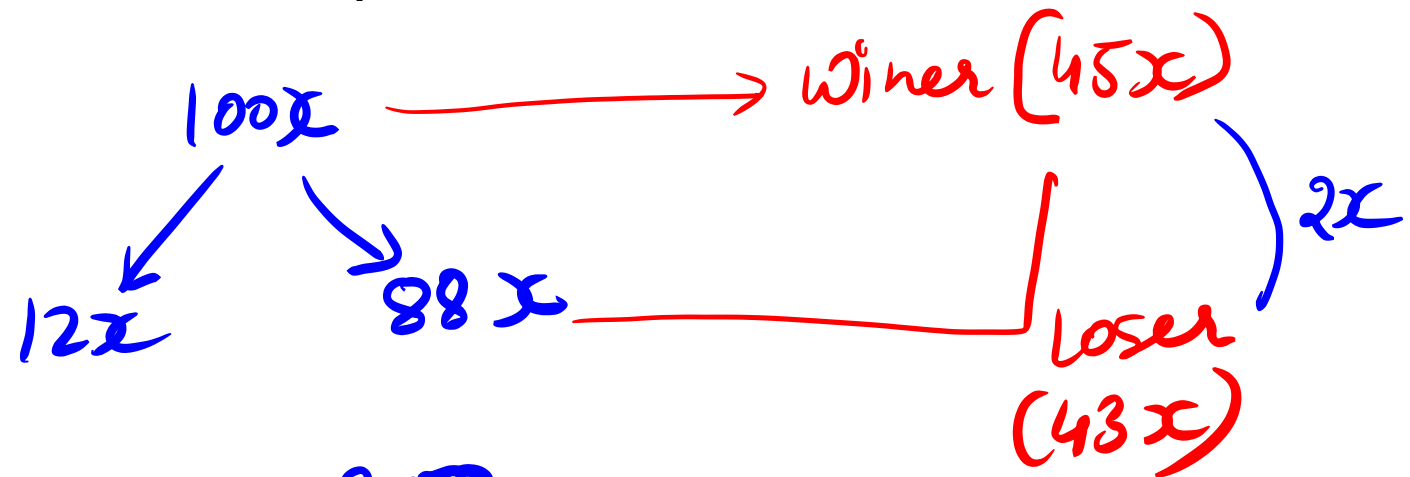
Question : In a election of 2 candidate 12 % of the voters did not cast their votes. The winner by getting 45% of the total votes, defeated his rival by 2000 votes. Find the total number of voters.

(a) 25000

(a) 50000

(a) 80000

(a) 100000



$$\begin{aligned} 2x &= 2000 \\ x &= 1000 \end{aligned}$$

$$\begin{aligned} \text{Total} &= 100x \\ &= 1000 \times 100 \\ &= \underline{\underline{1,00,000}} \end{aligned}$$



Question: In an Election, 10% persons didn't cast their votes and 10% found to be invalid. The winner got 54% of valid votes and won by 1620 votes. Find total number of votes?

(a) 12500
(b) 17500
(c) ~~25000~~
(d) 35000

Handwritten solution:

Let total number of votes be $100x$.

10% Not Casted $\rightarrow 10x$

90% Casted $\rightarrow 90x$

10% Invalid $\rightarrow 9x$ (Invalid)

81% (Valid) $\rightarrow 81x$

54% (winner) $\rightarrow 43.74x$

46% Loser $\rightarrow 37.26x$

8% difference $\rightarrow 4.48x$

Winner's votes: $43.74x$

Loser's votes: $37.26x$

Difference: $43.74x - 37.26x = 6.48x$

Given difference is 1620 votes.

$6.48x = 1620$

$x = 250$

Total votes: $100x = 25000$

Calculation shown in image:

$$81x \times \frac{8}{100} = 1620$$
$$x = 250$$
$$100x = 25000$$



Problem based on Population





Question : The population of a town is 50,000. It increases by 10% in the first year and 12% in the second year. What will be the population after 2 years.

(a) 55000

$$50000 \times \frac{110}{100} \times \frac{112}{100}$$

(a) 61600

~~61600~~

(a) 72700

(a) 84600



Question : The current population of a town is 28,000. During the last 2 years the population increased at the rate of 16% and 20% per year. The population 2 years ago was (approximately)

(a) 24000

(a) 22000

(a) 20000

(a) 18000

$$x \times \frac{116}{100} \times \frac{120}{100} = 28000$$

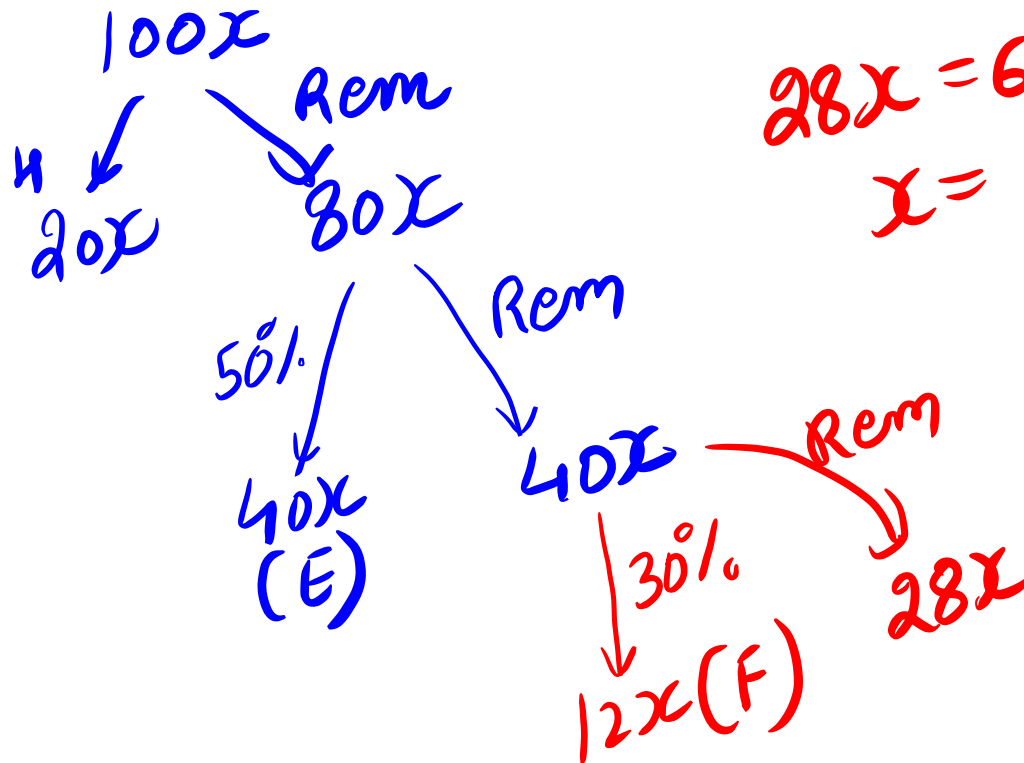
$$x = 20114$$

20000 Approx



Question: In a town 20% of the population are Hindi readers, 50% of the remaining are English readers and 30% of the remaining are French readers and rest 6300 people are regional language readers. Then find the total population in town.

- (a) 20000
- (b) ~~22500~~
- (c) 35000
- (d) 15000



$$28x = 6300$$

$$x = \frac{6300}{28} = 225$$

$$100x = 22500$$

